Course: ITAI 1371 – Fundamentals of Machine Learning

Lab 3: Reflection Journal

1. Architectural Clarity: From Code Execution to Lifecycle Thinking

Stepping through the 25 cells of Module_03_Lab_Exercise.ipynb shifted our perspective from viewing machine learning as isolated script executions to understanding it as an interconnected, deterministic data lifecycle. Earlier in the course, Python packages like Pandas, NumPy, and Scikit-Learn felt like utility tools for tabular cleanup. In this lab, their programmatic synchronization revealed the true ML pipeline: **Data Exploration -> Preprocessing & Feature Selection -> Deterministic Splitting -> Model Induction -> Quantitative Evaluation -> Interpretation**.

The initial barrier was understanding what happens under the hood during .fit() versus .predict(). It is easy to treat these calls as black boxes, but tracking feature transformations highlight how parameter weights and decision boundaries are mathematically established. In supervised learning, the objective function iteratively updates internal parameters using labeled target vectors (y). Conversely, inference through .predict() isolates those learned coefficients and applies them blindly to unseen matrices (Xtest). Realizing that data leakage occurs the moment test data contaminates preprocessing boundaries makes the rigid separation of steps intuitive rather than arbitrary.

2. Paradigm Distinctions: Supervised, Unsupervised, and Reinforcement Learning

Analyzing the three fundamental learning paradigms resolved ambiguities regarding their operational boundaries and training inputs:

- **Supervised Learning:** Defined strictly by the presence of a ground-truth supervisory signal (y). Whether addressing discrete outputs in classification (e.g., cultivars in the Wine dataset or malignant/benign diagnoses in Breast Cancer Wisconsin) or continuous values in regression, the algorithm adjusts itself relative to explicit residual loss. The key mental model is having an immutable "answer key."
- **Unsupervised Learning:** Operates entirely without labels (X only). Instead of mapping inputs to predefined targets, algorithms discover intrinsic geometric structures, cluster centroids, or latent manifold representations. While supervised models predict, unsupervised models describe structure and density.
- **Reinforcement Learning (RL):** Contrasts sharply with both static paradigms. In RL, there is no static dataset; an agent interacts dynamically with an environment via

action-state transitions. Optimization is driven by sparse, delayed scalar rewards rather than sample-by-sample corrections.

Categorizing scenarios in Task 8 demonstrated why problem formulation dictates algorithm selection: housing price estimation requires mapping features directly to numeric labels (Supervised Regression), while market segmentation requires spatial proximity clustering across unlabeled records (Unsupervised Learning).

3. Empirical Analysis: The Wine Classification Pipeline

Working with the Italian Wine dataset (178 samples, 13 chemical features across 3 cultivars) provided concrete evidence of how model architecture influences predictive boundaries.

Initial Baseline (4 Features)	-->	Optimized Feature Subset (3 Features)
['alcohol', 'malic_acid', 'ash', 'alcalinity']		['alcohol', 'hue', 'flavanoids']
Logistic Regression: 88.9% Acc		Logistic Regression: 94.4% Acc
Decision Tree (depth=3): 83.3% Acc		

Algorithmic Behavior Comparison:

1. **Logistic Regression (88.9% Baseline):** Utilized a linear combination of input features passed through a multinomial softmax function. It achieved a balanced Macro F1-score of 0.88, demonstrating that continuous chemical concentrations frequently exhibit linearly separable trends in higher-dimensional hyperplanes.
2. **Decision Tree (83.3% Baseline):** Constrained to an orthogonal axis-aligned partition at max_depth=3. The tree struggled particularly on Class 1 and Class 2 separation, dropping recall on Class 1 to 71% (0.80 F1-score). Because decision trees split along single-feature orthogonal thresholds, they require deeper branch complexity or oblique splits to match the continuous boundaries that linear models draw effortlessly across correlated continuous chemical data.

The Power of Feature Selection (Task 1 Breakthrough): When Marisabel modified the feature subspace from the arbitrary first four attributes to ['alcohol', 'hue', 'flavanoids'], the test accuracy jumped from **88.9% to 94.4%**. Examining the initial EDA correlation heatmaps explained this gain: flavanoids and hue show strong separation power across cultivar pigment profiles, whereas ash and alcalinity_of_ash exhibited collinear noise with low discriminatory variance. This exercise proved that feeding raw or redundant features degrades model generalization. Thoughtful feature selection consistently outperforms arbitrary dimensionality expansion.

4. Generalization & Diagnostic Metrics: Beyond Accuracy

The lab highlighted why high aggregate accuracy can conceal systemic failure modes. Evaluating the baseline Logistic Regression via the multi-class Confusion Matrix and Classification Report provided critical diagnostic depth:

Precision=TP+FPTPRecall=TP+FNTP

While the model achieved an overall accuracy of 88.9%, examining Class 2 revealed a recall drop to **0.70** (only 7 out of 10 samples correctly identified), compared to a perfect **1.00** recall for Class 0. Relying strictly on accuracy would obscure the fact that nearly a third of Class 2 wines were misclassified as Class 1.

Applying stratify=y during train_test_split proved essential. Because the Wine dataset contains an uneven class distribution (71 Class 1, 59 Class 0, 48 Class 2), an unstratified split risks test-set sampling bias, where a minority class could be severely underrepresented, destabilizing the confusion matrix metrics.

5. Transfer Learning of Concepts: The Breast Cancer Practice Test

Applying the end-to-end pipeline to the Breast Cancer Wisconsin dataset (569 samples, 30 features, binary target) verified that the ML workflow is universally portable across domains:

Breast Cancer Wisconsin Workflow (Logistic Regression, max_iter=10000)

```
| Shape: (569, 30)       | ──> | Train: 455 | Test: 114 | ──> | Test Accuracy: 95.6%  |
| Target: [Malig, Benign] |     | Split: 80/20 Stratified |     | Malignant Recall: 0.91 |
```

The model attained **95.6% accuracy**, but clinical domain requirements shifted the group's analytical focus entirely to **Recall on the Malignant class (0.91)**. A False Positive (predicting malignant when benign, lowering precision to 0.97) results in secondary testing, whereas a False Negative (predicting benign when malignant, lowering recall to 0.91) delays critical medical intervention.